MIL-DTL-44033B
21 April 1995
SUPERSEDING.
~~MIL-L-44033A~~
25 October 1984

DETAIL SPECIFICATION

LINES, MULTI-LOOP, CONCENTRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of concentric multi-loop lines fabricated of webbing and intended for use in airdrop operations.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are needed to meet the requirements specified in sections 3, 4, and 5 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that-they must meet all requirements documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, Engineering Center, Attn: SSCNC-UC, Natick, MA 01760-5017 by using the Standardization Document Improvement Proposal (DD form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1670

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2.2 Government documents, drawings, and publications. The following Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

11-1-3164 - Riser Extension Assembly
68F217 - Line MultiLoop, Concentric Loop Type, Assembly of

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, Attn: SATNC-MI, Natick, MA, 01760-5017.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS [ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Samples. The color shade of webbing specified on the drawings shall match the standard shade sample (see 6.3).

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3.3 Materials. The materials used in the manufacture of the lines shall conform in all aspects to the requirements specified on the drawings and parts lists pertaining thereto, and as specified herein. Recovered materials shall be used to the maximum extent possible.

3.4 Construction. Construction shall conform to all aspects of the requirements specified on the drawings and as specified herein. Repairs to defective stitching shall be as specified herein. Other repairs or rework shall not be performed on the lines.

3.4.1 Webbing relaxation. All webbing used in the fabrication of loops shall be relaxed for a minimum of 48 hours prior to measuring for cutting. Relaxation of the webbing shall be accomplished by loosening the coiled rings so there is a minimum of 1/4 inch space between the webbing coils of the roll.

3.4.2 Dimensions. Finished lengths and the number of loops for the lines shall be as specified in the contract (see 6.2). The cut length of the individual loops, after relaxation of the webbing rolls (see 3.4.1) and prior to splicing, shall be determined by the contractor and shall result in the finished length shown on the drawings.

3.4.3 Machine stitching. Ends of stitching shall be back stitched not less than 1/2 inch. Thread tension shall be maintained so there will be no loose stitching and the locks shall be embedded into the materials sewn. all thread ends shall be trimmed so that they do not exceed 1/4 inch.

3.4.3.1 Automactic Stitching. Automatic sewing machines may be used to perform any of the stitch patterns provided the requirements for the stitch pattern, stitches per inch, size and type of thread are met; and at no less than two tying, overlapping or backstitches are used to secure the ends of the stitching.

3.4.3.2 Loose stitches. Looses stitches shall not be removed but shall be reinforced by adjacent stitching.

3.4.3.2 Tight stitches. Tight stitches which occur on the sliding keeper only shall be removed and the section shall be restitched.

3.4.3.4 Repairs of the 301 stitching. Repairs of type 301 stitching shall be as follows:

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a. When thread breaks, skipped stitches, run-offs, or bobbin runouts occur during sewing, the stitching shall be repaired by adjacent stitching with the stitches extending a minimum of 1 inch beyond the ends of the stitches being repaired. When making the above repairs, the ends of stitching are not required to be backstitched.

b. Thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by adjacent stitching. The stitching shall start a minimum of 1 inch in back of the defective area, and continue a minimum of 1 inch beyond the ends of the stitching being repaired. When making the above repairs, the ends of stitching are not required to be backstitched.

3.5 Identification marking. Using the marking medium specified on the drawing, all marking shall be of characters 1/2 inches in height. Markings shall be located near each end of the line of one side, and shall be placed so that they will not be obscured by the keepers. No marking shall appear on splices or keepers. Identification shall be as follows:

a. US
b. Part No.
c. Finished length for P/N OF Drawing 68F217 only
d. Date of manufacture
e. Manufacturer's name or code

3.6 Workmanship. Concentric lines shall be uniform in quality and shall be free from irregularities or defects that could adversely affect performance or durability.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one complete concentric line assembly when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.4, 4.5 and 4.6.

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.5 and 4.6.

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4.4 Examination. Each concentric line assembly shall be examined for compliance with the requirements specified in 3.2 though 3.5. This element of inspection shall encompass all visual examinations (see 4.5) and dimensional measurements (see 4.6). Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection of the lot.

4.5 End item visual examination. The end items shall be examined for the defects listed in Table I. Noncompliance with any specified requirement shall constitute a defect.

TABLE I. End item visual examination

Examine	Defect	Classification major	Classification minor
Webbing	Not specified type or class	101	
	Color not as specified		201
	Ends not wax dipped or seamed as specified		202
	Hole, tear, or cut in material	102	
	Abrasion resulting in rupture of yarns, or in nap sufficient to obscure the identity of any yarns exceeding 10 percent of width or 1 inch in length	103	
	Edge frayed, beaded, corded, or torn	104	
	Two broken warp or fill yarns		203
	More than two broken warp or fill yarns	105	
Open Seams	1/2 inch or less		204
	More than 1/2 inch	106	
	Note: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or runoff stitches occur.		
Run-off	(See Open Seams)		
Stitch type	Wrong stitch type	107	

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TABLE I. End item visual examination (cont'd)

Examine	Defect	Classification major	Classification minor
Stitch tension	Loose resulting in loose top or bobbin thread		205
	Excessively tight resulting in puckering of webbing (applicable to sliding keepers only)		206
Stitches per inch	One stitch per inch more than maximum specified		207
	More than one stitch per inch more than maximum specified	108	
	One stitch per inch less than minimum specified		208
	More than one stitch per inch less than minimum specified	109	
	NOTE: Variation in the number of stitches per inch caused by the operator speeding up machine to sew over heavy places, shall be classified as follows: (a) Within the major defect classification minor defect (b) Within the minor defect classification - no defect		
Thread breaks, skipped stitches, or runoffs	Not adjacent stitched as specified		209
	NOTE: Thread breaks or two or more consecutive skipped or runoff stitches not adjacent stitched shall be classified as open seams.		
Ends of stitching	Not secured as specified		210
Rows of stitching	Any row of stitching missing, except on 72 continuous lines and six point stitch patterns	110	
	On 72 continuous lines of stitching one row more or less than specified	111	

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TABLE I. End item visual examination (cont'd)

Examine	Defect	Classification major	Classification minor
	On six point stitch pattern, more or less than one row specified per individual stitch pattern	112	
Components and Assembly	Any component part omitted or not as specified or required operation omitted (unless otherwise classified herein)	113	
	Any component misplaced or not assembled as specified	114	
	Needle chew		211
	Any darn, splice or other unauthorized repair	115	
Sliding Keeper	Not specified size or type	116	
	Sliding keeper not completely dipped in melted wax mixture		212
	Stitch pattern on overlap of keepers not in location shown on Drawing 68F217 in relation to top or bottom loops		213
	One or both sliding keepers omitted	117	
Fixed Keeper	Not specified size or type	118	
	Not assembled as specified		214
	One or more fixed keepers omitted	119	
Splices	Not spaced alternately as specified		215
	Not joined parallel as specified	120	
	More than two splices per loop	121	
Buffer	Not specified size or type	122	
	Not assembled as specified		216
	One or both buffers omitted	123	
	Not longitudinally stitched to loop	124	

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TABLE I. End item visual examination (cont'd)

Examine	Defect	Classification major	Classification minor
Cleanness	Dirt, grease, oil or foreign matter clearly noticeable		217
Identification marking ,	Missing, incomplete, not legible, improper location or size		218
	Wrong color		219

4.6 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect.

5. PACKAGING

5.1 Preservation and packing. Packaging requirements shall be as specified in the contract or order (see 6.2) .

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with ASTM D 3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The items are intended for use in the rigging, extraction and delivery of cargo and equipment from military aircraft in flight.

6.2 Agusition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).
c. Stock number.
d. When a first article is required (see 3.1).
e. Finished length and number of loops (3.4.2).

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6.3 Sample. For access to government furnished property (see 3.2) address the contracting activity issuing the invitation for bids or request for proposal.

6.4 International standardization agreement. Design provisions of this document are the subject of international standardization agreements NATO STANAG No. 3469 (Edition No. 2) and ASCC AIR STD 44/15E. When amendment, revision, or cancellation of this document is proposed which will affect or violate the international agreement concerned the preparing activity will take appropriate reconciliation action though international standardization channels including Departmental Standardization Offices., if required.

6.5 Subject term (key word) listing.

Air delivery	Parachute
Extraction	Rigging

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - GL
Navy - AS
Air Force - 99

Review activities:
Army - AV
Navy - MC
Air Force - 82

Preparing activity:
Army - GL

(Project 1670-0825)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-44033B	2. DOCUMENT DATE (YYMMDD) 950421

3. DOCUMENT TITLE
LINES, MULTI-LOOP, CONCENTRIC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME U.S. Army Natick RD&E Center	b. TELEPHONE (Include Area Code) (1) Commercial: (508) 651-4258 (2) AUTOVON/DSN: 256-4258
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SSCNC-UC Natick, MA 01760-5017	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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